HARD

PROBLEM-SOLVING AND DATA ANALYSIS

Two-variable data: Models and scatterplots

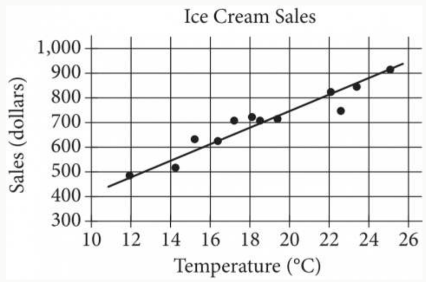
01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS



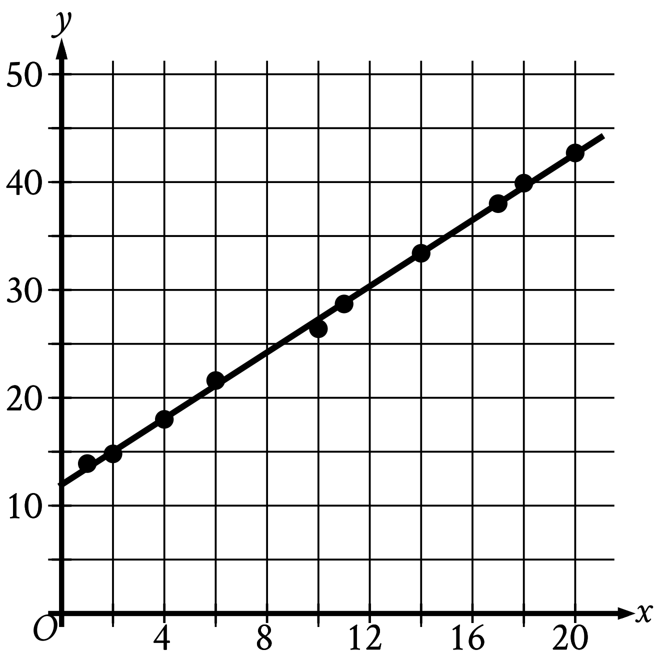
The scatterplot above shows a company's ice cream sales d, in dollars, and the high temperature t, in degrees Celsius (°C), on 12 different days. A line of best fit for the data is also shown. Which of the following could be an equation of the line of best fit?

A. $d = 0.03t + 402$

B. $d = 10t + 402$

C. $d = 33t + 300$

D. $d = 33t + 84$



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending up from left to right.
- A line of best fit is shown:
    - The line of best fit slants up from left to right.
    - The line of best fit passes through the following approximate coordinates:
        - (0 comma 11.9)
        - (12 comma 30.3)
        - (20 comma 42.6)

The scatterplot shows the relationship between two variables, x and y, for data set E. A line of best fit is shown. Data set F is created by multiplying the y-coordinate of each data point from data set E by 3.9. Which of the following could be an equation of a line of best fit for data set F?

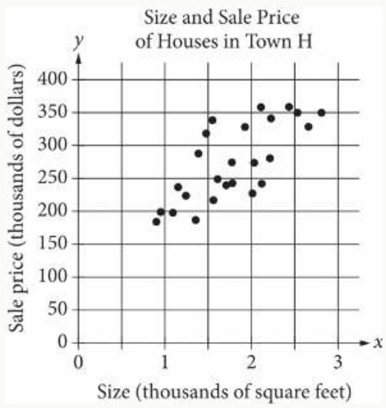
A. $y = 46.8 + 5.9x$

B. $y = 46.8 + 1.5x$

C. $y = 12 + 5.9x$

D. $y = 12 + 1.5x$

Q3 PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the size x and the sale price y of 25 houses for sale in Town H. Which of the following could be an equation for a line of best fit for the data?

A. $y = 200x + 100$

B. $y = 100x + 100$

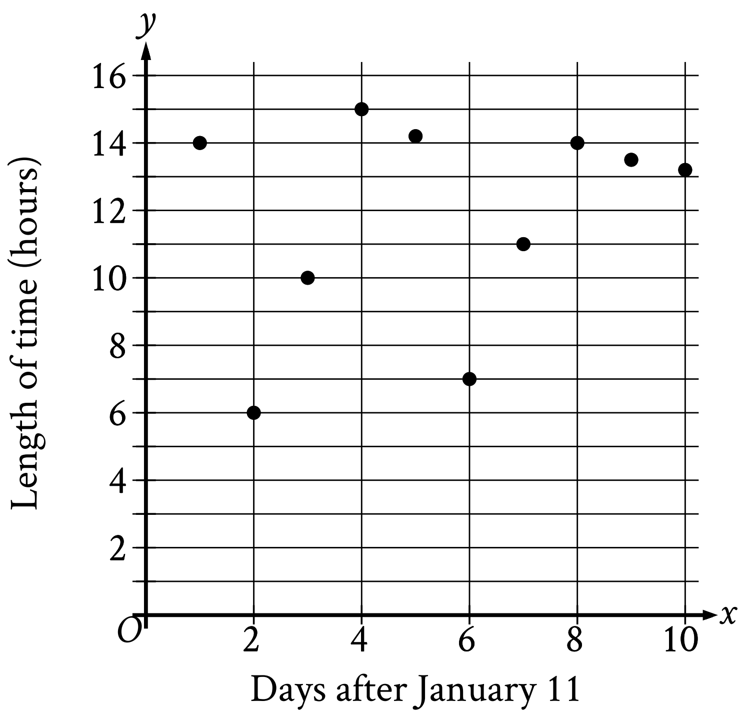
C. $y = 50x + 100$

D. $y = 100x$

Q4

GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS

The scatterplot shows the relationship between the length of time y, in hours, a certain bird spent in flight and the number of days after January 11, x.



- The scatterplot has 10 data points.
- The data points are spread out.
- The data points have the following coordinates:
  - (1 comma 14)
  - (2 comma 6)
  - (3 comma 10)
  - (4 comma 15)
  - (5 comma 14.2)
  - (6 comma 7)
  - (7 comma 11)
  - (8 comma 14)
  - (9 comma 13.5)
  - (10 comma 13.2)

What is the average rate of change, in hours per day, of the length of time the bird spent in flight on January 13 to the length of time the bird spent in flight on January 15?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 PROBLEM-SOLVING AND DATA ANALYSIS

For $x > 0$, the function f is defined as follows:

$$f(x) \text{ equals } 201\% \text{ of } x$$

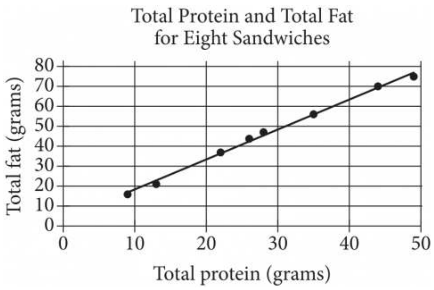
Which of the following could describe this function?

A. Decreasing exponential

B. Decreasing linear

C. Increasing exponential

D. Increasing linear



The scatterplot above shows the numbers of grams of both total protein and total fat for eight sandwiches on a restaurant menu. The line of best fit for the data is also shown. According to the line of best fit, which of the following is closest to the predicted increase in total fat, in grams, for every increase of 1 gram in total protein?

A. 2.5

B. 2.0

C. 1.5

D. 1.0

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

	Amount invested	Balance increase
Account A	\$500	6% annual interest
Account B	\$1,000	\$25 per year

Two investments were made as shown in the table above. The interest in Account A is compounded once per year. Which of the following is true about the investments?

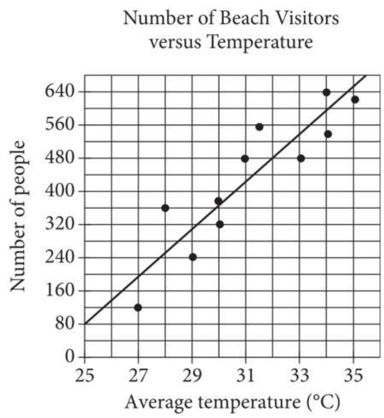
A. Account A always earns more money per year than Account B.

B. Account A always earns less money per year than Account B.

C. Account A earns more money per year than Account B at first but eventually earns less money per year.

D. Account A earns less money per year than Account B at first but eventually earns more money per year.

Q8 GRID-IN PROBLEM-SOLVING AND DATA ANALYSIS



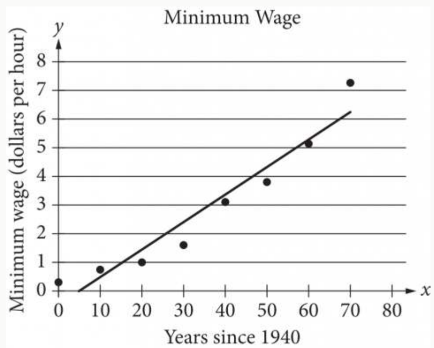
Each dot in the scatterplot above represents the temperature and the number of people who visited a beach in Lagos, Nigeria, on one of eleven different days. The line of best fit for the data is also shown. The line of best fit for the data has a slope of approximately 57. According to this estimate, how many additional people per day are predicted to visit the beach for each 5°C increase in average temperature?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

PROBLEM-SOLVING AND DATA ANALYSIS



The scatterplot above shows the federal-mandated minimum wage every 10 years between 1940 and 2010. A line of best fit is shown, and its equation is $y = 0.096x - 0.488$. What does the line of best fit predict about the increase in the minimum wage over the 70-year period?

A. Each year between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

B. Each year between 1940 and 2010, the average increase in minimum wage was 0.49 dollars.

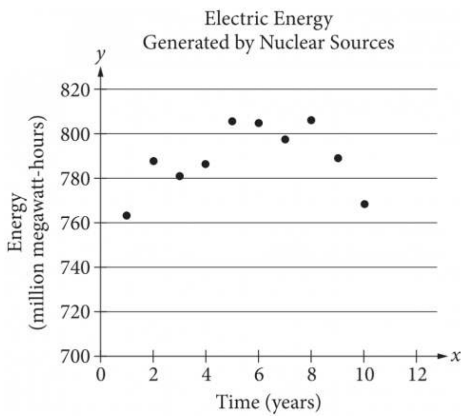
C. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.096 dollars.

D. Every 10 years between 1940 and 2010, the average increase in minimum wage was 0.488 dollars.

Q10

PROBLEM-SOLVING AND DATA ANALYSIS

The scatterplot below shows the amount of electric energy generated, in millions of megawatt-hours, by nuclear sources over a 10-year period.



Of the following equations, which best models the data in the scatterplot?

A. $y = 1.674x^2 + 19.76x - 745.73$

B. $y = -1.674x^2 - 19.76x - 745.73$

C. $y = 1.674x^2 + 19.76x + 745.73$

D. $y = -1.674x^2 + 19.76x + 745.73$